Q1. Python Libraries - NumPy

Original Aim: To use NumPy for basic numerical operations on the Iris dataset. Load the Iris data and select a numerical attribute such as SepalLengthCm. Convert its values into a NumPy array and calculate the sum, mean, maximum and minimum. Also display a slice containing the first five values.

Aim

Use NumPy for basic numerical operations on SepalLengthCm and display its first five values.

Requirements

Python, NumPy, Pandas and scikit-learn; the bundled Iris dataset. See references [D1] and [D2].

Essential theory

A NumPy ndarray is a multidimensional numerical container with a consistent data type. Here one measurement column becomes a one-dimensional floating-point array of length 150. Vectorized aggregations operate on the array without manually writing a Python loop. Python uses zero-based indexing; a slice's stop position is excluded.

Step-by-step procedure

1. Load Iris and construct the five-column DataFrame with the assigned names.
2. Select SepalLengthCm and convert it with to_numpy().
3. Apply sum, mean, max and min to the array.
4. Display values[:5], which selects positions 0 through 4.

Complete Python program

```
import numpy as np
from sklearn.datasets import load_iris
import pandas as pd

iris = load_iris()
features = [
    "SepalLengthCm", "SepalWidthCm",
    "PetalLengthCm", "PetalWidthCm"
]
species = ["Iris-" + name for name in iris.target_names]
df = pd.DataFrame(iris.data, columns=features)
df["Species"] = pd.Categorical.from_codes(
    iris.target, categories=species
)

values = df["SepalLengthCm"].to_numpy()
print("NumPy array shape:", values.shape)
print(f"Sum: {np.sum(values):.6f}")
print(f"Mean: {np.mean(values):.6f}")
print(f"Maximum: {np.max(values):.6f}")
print(f"Minimum: {np.min(values):.6f}")
print("First five values:", values[:5])
```

Code explanation

- `load_iris()` provides features and integer targets. `DataFrame(..., columns=features)` maps measurements to the exact required names.
- `pd.Categorical.from_codes` converts target codes into meaningful species labels; Species is not used in these calculations.

- `to_numpy()` obtains the selected column's array. Its shape is (150,), meaning one dimension with 150 values.
- `np.sum` adds all values; `np.mean` divides the total by the number of values; `np.max` and `np.min` return the extreme values.
- `values[:5]` takes five elements, not six. `.6f` controls displayed decimal places only.

Verified output

```
NumPy array shape: (150,)
Sum: 876.500000
Mean: 5.843333
Maximum: 7.900000
Minimum: 4.300000
First five values: [5.1 4.9 4.7 4.6 5. ]
```

Observation

The sepal lengths range from 4.3 to 7.9 cm, with a mean of about 5.8433 cm across 150 flowers. The first-five slice preserves the original dataset order; it is not a sorted sample. The arithmetic sum is 876.5, but it is mainly a demonstration of aggregation rather than a biologically useful measurement.

Result

The requested numerical column was converted to a NumPy array, summarized with four aggregation functions and sliced successfully.

Common errors and troubleshooting

- NameError for np: include `import numpy as np` in the same fresh session.
- KeyError: check exact capitalization and the Cm suffix in SepalLengthCm.
- Wrong first-five result: use [:5], not [1:5], which returns only four values.

Viva questions and answers

- What is ndarray? NumPy's N-dimensional array type, used here as a one-dimensional numeric array.
- What is vectorization? Applying an operation to array data without an explicit Python element-by-element loop.
- Is the stop index included in a slice? No; [:5] includes positions 0, 1, 2, 3 and 4.
- Does rounding the printout change the array? No; the f-string changes presentation only.